

## APATITY, Russian Federation

WMO: 222130

Lat: 67.5508N

Lon: 33.3622E

Elev: 439

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p) |
| 1             | -25.1      | -20.1 | -29.2                         | 1.1 | -24.7 | -24.1 | 1.5 | -20.2 | 22.6                  | 19.9 | 20.5 | 22.3 | 0.8                      | 100  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.2                   | 75.1            | 61.8 | 71.7 | 59.9 | 67.8 | 58.0 | 63.7                | 72.1 | 61.1 | 69.0 | 59.2 | 66.2 | 6.7                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 59.8                            | 77.9 | 67.6 | 57.6 | 72.1 | 64.9 | 55.8 | 67.3 | 62.8 | 29.2          | 72.8 | 27.3 | 68.9 | 26.0 | 66.6 | 70.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 18.8          | 16.4 | 14.3 | DB                         | -29.3 | 79.6               | 4.0 | 5.0                                                | -32.2 | 83.2       | -34.6 | 86.2       | -36.8 | 89.0       | -39.8 | 92.6 | (4) |
| (5)               |      |      | WB                         | -30.2 | 66.1               | 4.3 | 3.3                                                | -33.3 | 68.5       | -35.8 | 70.4       | -38.3 | 72.2       | -41.4 | 74.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 32.1        | 6.7   | 13.6  | 19.2  | 29.4  | 41.6  | 50.4  | 57.7  | 53.7  | 44.7  | 32.3  | 21.6  | 13.1  |  |  |
|                                                                        | DBStd       | 19.31       | 14.10 | 13.22 | 10.45 | 8.10  | 7.71  | 7.29  | 6.54  | 6.05  | 5.74  | 8.02  | 10.60 | 13.49 |  |  |
|                                                                        | HDD50       | 7052        | 1342  | 1018  | 955   | 617   | 280   | 82    | 10    | 30    | 175   | 548   | 851   | 1144  |  |  |
|                                                                        | HDD65       | 12023       | 1807  | 1438  | 1420  | 1067  | 727   | 441   | 238   | 354   | 608   | 1013  | 1301  | 1609  |  |  |
|                                                                        | CDD50       | 523         | 0     | 0     | 0     | 0     | 18    | 94    | 250   | 144   | 17    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 20          | 0     | 0     | 0     | 0     | 0     | 3     | 13    | 4     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 140         | 0     | 0     | 0     | 0     | 2     | 19    | 100   | 19    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 9           | 0     | 0     | 0     | 0     | 0     | 0     | 8     | 1     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.3         | 6.4   | 6.2   | 6.5   | 6.8   | 7.0   | 6.9   | 6.2   | 5.7   | 6.3   | 6.4   | 6.0   | 5.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 29.9        | 2.6   | 2.1   | 2.1   | 1.6   | 2.0   | 2.6   | 3.0   | 2.8   | 2.6   | 3.1   | 2.6   | 2.8   |  |  |
|                                                                        | PrecMax     | 34.7        | 4.1   | 3.6   | 4.1   | 3.7   | 3.2   | 4.3   | 4.7   | 4.6   | 4.4   | 5.8   | 4.6   | 4.5   |  |  |
|                                                                        | PrecMin     | 25.9        | 1.0   | 0.9   | 1.2   | 0.5   | 1.0   | 0.8   | 1.2   | 1.1   | 1.1   | 1.1   | 1.1   | 1.7   |  |  |
|                                                                        | PrecStd     | 2.4         | 0.7   | 0.7   | 0.7   | 0.6   | 0.6   | 0.9   | 0.8   | 0.9   | 0.8   | 1.1   | 0.8   | 0.7   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 36.6  | 36.5  | 40.5  | 53.1  | 69.9  | 76.5  | 80.9  | 76.2  | 62.2  | 49.2  | 38.5  | 35.5  |  |  |
|                                                                        |             | MCWB        | 34.2  | 34.9  | 35.0  | 43.3  | 54.8  | 60.1  | 65.8  | 65.4  | 55.2  | 46.4  | 36.6  | 34.0  |  |  |
|                                                                        | 2%          | DB          | 31.1  | 34.1  | 36.3  | 45.9  | 63.7  | 72.1  | 76.5  | 71.1  | 58.5  | 46.2  | 36.4  | 32.8  |  |  |
|                                                                        |             | MCWB        | 29.5  | 33.0  | 32.8  | 38.8  | 51.5  | 58.8  | 62.2  | 61.0  | 53.0  | 43.8  | 35.5  | 31.7  |  |  |
|                                                                        | 5%          | DB          | 28.6  | 32.6  | 34.2  | 42.1  | 58.9  | 67.4  | 73.6  | 66.5  | 55.8  | 44.4  | 34.9  | 31.0  |  |  |
|                                                                        |             | MCWB        | 27.5  | 31.8  | 31.7  | 37.1  | 49.1  | 55.9  | 61.3  | 59.0  | 51.6  | 42.5  | 34.1  | 30.1  |  |  |
|                                                                        | 10%         | DB          | 26.0  | 30.5  | 32.3  | 38.9  | 54.1  | 63.3  | 70.2  | 63.3  | 53.2  | 42.3  | 33.3  | 29.2  |  |  |
|                                                                        |             | MCWB        | 25.1  | 29.5  | 30.5  | 34.7  | 46.3  | 53.8  | 59.6  | 57.2  | 49.7  | 40.6  | 32.6  | 28.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 34.1  | 35.0  | 35.2  | 43.5  | 56.2  | 62.8  | 66.9  | 67.1  | 56.6  | 47.2  | 37.2  | 34.2  |  |  |
|                                                                        |             | MCDB        | 36.2  | 35.8  | 39.2  | 51.0  | 67.4  | 72.6  | 76.8  | 75.3  | 59.9  | 48.4  | 37.8  | 35.2  |  |  |
|                                                                        | 2%          | WB          | 29.9  | 33.1  | 33.5  | 39.7  | 52.6  | 59.9  | 64.9  | 62.3  | 53.8  | 44.3  | 35.8  | 32.0  |  |  |
|                                                                        |             | MCDB        | 30.7  | 33.8  | 35.8  | 45.4  | 61.5  | 69.8  | 74.3  | 69.3  | 57.3  | 45.7  | 36.4  | 32.5  |  |  |
|                                                                        | 5%          | WB          | 27.5  | 31.8  | 32.1  | 37.0  | 49.7  | 57.0  | 62.5  | 59.7  | 52.0  | 42.7  | 34.1  | 30.2  |  |  |
|                                                                        |             | MCDB        | 28.5  | 32.6  | 33.8  | 41.1  | 58.8  | 65.0  | 70.0  | 65.1  | 55.1  | 44.4  | 34.6  | 30.9  |  |  |
|                                                                        | 10%         | WB          | 25.2  | 29.4  | 30.7  | 35.6  | 46.7  | 54.9  | 60.3  | 57.8  | 50.1  | 40.7  | 32.9  | 28.4  |  |  |
|                                                                        |             | MCDB        | 26.1  | 30.2  | 32.2  | 38.1  | 53.8  | 62.5  | 68.3  | 62.6  | 52.7  | 42.3  | 33.4  | 29.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 14.2  | 11.9  | 14.6  | 13.1  | 13.0  | 13.3  | 14.2  | 12.2  | 10.8  | 7.9   | 9.3   | 13.3  |  |  |
|                                                                        |             | MCDBR       | 12.2  | 7.6   | 12.6  | 14.6  | 22.5  | 20.6  | 22.8  | 17.9  | 16.1  | 8.5   | 6.7   | 10.0  |  |  |
|                                                                        | 5% WB       | MCWBR       | 11.9  | 7.5   | 9.7   | 9.5   | 12.7  | 10.3  | 10.9  | 9.7   | 10.5  | 7.0   | 6.4   | 9.2   |  |  |
|                                                                        |             | MCWBR       | 11.9  | 8.3   | 10.7  | 11.8  | 21.8  | 17.2  | 17.8  | 16.2  | 13.2  | 7.7   | 6.5   | 9.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.215 | 0.271 | 0.309 | 0.344 | 0.361 | 0.359 | 0.379 | 0.358 | 0.326 | 0.288 | N/A   | N/A   |  |  |
|                                                                        |             | taud        | 1.906 | 2.030 | 2.068 | 2.120 | 2.303 | 2.420 | 2.396 | 2.428 | 2.482 | 2.252 | N/A   | N/A   |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 33    | 162   | 225   | 252   | 263   | 268   | 257   | 250   | 230   | 162   | N/A   | 0     |  |  |
|                                                                        |             | Edh at Noon | 4     | 17    | 29    | 36    | 34    | 31    | 31    | 27    | 20    | 14    | N/A   | 0     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 15    | 157   | 594   | 1117  | 1445  | 1604  | 1459  | 1013  | 548   | 177   | 31    | 0     |  |  |
|                                                                        | RadStd      | RadStd      | 1     | 24    | 57    | 89    | 117   | 151   | 183   | 93    | 48    | 26    | 3     | 0     |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page